span_corruption.py

`src/moe_pruning/data/span_corruption.py · 104 lines · module moe_pruning.data.span_corruption`

What it does

Deterministic T5 span corruption used to evaluate switch-base-8 on its pre-training objective: parts of a document are replaced by sentinel tokens and the model must reproduce the removed spans.

How it works

- `random_spans_noise_mask()` is a NumPy port of the T5 reference implementation (noise density, mean span length).
- Each example uses its own random generator seeded with `(EVAL_SEED, example_id)`, so masks do not depend on which other examples are evaluated.
- `replace_spans_with_sentinels()` builds the corrupted input (`<extra_id_N>` per removed span) and, with the inverted mask, the target.
- `corrupt_example()` truncates the document so both input and target fit the configured lengths.

Notes

- The reported loss on these targets is a denoising loss, not perplexity.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`random_spans_noise_mask(length, noise_density, mean_noise_span_length, rng)`	Boolean mask of `length` with ~noise_density noise in spans of mean length.
`sentinel_ids(tokenizer, count)`	Token ids of the T5 sentinels <extra_id_0> ... <extra_id_99>.
`replace_spans_with_sentinels(tokens, mask, sentinels)`	Replace each masked span by one sentinel (in order) and drop the rest of the span.
`class CorruptedExample`	One span-corruption example: source, corrupted input, target, noise mask.
`corrupt_example(example_id, token_ids, tokenizer, noise_density, mean_noise_span_length, seed, max_source_length, max_target_length)`	Corrupt one document deterministically and fit it into the length limits.

Connections

Used by: `src/moe_pruning/tasks/seq2seq_denoising.py`